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TURBINE VANE WITH LEADING EDGE COOLING

Abstract

A vane for a gas turbine engine, including: an airfoil extending between an inner platform and an outer platform, the outer platform of the vane having a leading edge and a trailing edge; a passage located in the vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of passages each of the plurality of passages fanning outwardly from the passage and each of the plurality of passages being separated by one of a plurality of ribs located in the outer platform.

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Background/Summary

BACKGROUND

[0002] This disclosure relates to a gas turbine engine, and more particularly to a turbine vane with cooling features.

[0003] Gas turbine engines typically include a compressor section, a combustor section and a turbine section. During operation, air is pressurized in the compressor section and is mixed with fuel and burned in the combustor section to generate hot combustion gases. The hot combustion gases are communicated through the turbine section, which extracts energy from the hot combustion gases to power the compressor section and other gas turbine engine loads.

[0004] Both the compressor and turbine sections may include alternating series of rotating blades and stationary vanes that extend into the core flow path of the gas turbine engine. For example, in the turbine section, turbine blades rotate and extract energy from the hot combustion gases that are communicated along the core flow path of the gas turbine engine. The turbine vanes, which generally do not rotate, guide the airflow and prepare it for the next set of blades.

[0005] As mentioned above, hot combustion gases are communicated through the turbine section, which causes the turbine vanes to be exposed to high temperatures. As such, it is desirable to provide cooling to the turbine vanes.

BRIEF DESCRIPTION

[0006] Disclosed is a vane for a gas turbine engine, including: an airfoil extending between an inner platform and an outer platform, the outer platform of the vane having a leading edge and a trailing edge; a passage located in the vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of passages each of the plurality of passages fanning outwardly from the passage and each of the plurality of passages being separated by one of a plurality of ribs located in the outer platform.

[0007] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum has a larger volume than each one of the plurality of passages.

[0008] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, cooling openings extending from the internal plenum to an external surface of the leading edge.

[0009] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, cooling openings extending from the internal plenum to any one of an external surface of the leading edge, matefaces of the outer platform, and a radially inwardly facing surface of the outer platform.

[0010] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum is located between the plurality of passages and an external surface of the outer platform.

[0011] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an area of each of the passages is equal.

[0012] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, each of the plurality of passages are connected to the internal plenum.

[0013] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the vane is a turbine vane.

[0014] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the vane is formed from a casting process.

[0015] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, upper surfaces and/or lower surfaces of the plurality of passages have trip strips extending therefrom.

[0016] Also disclosed is a gas turbine engine, including: a fan section; a compressor section; a

combustor section; and a turbine section, the turbine section including at least one vane, the at least one vane including: an airfoil extending between an inner platform and an outer platform, the outer platform of the at least one vane having a leading edge and a trailing edge; a passage located in the at least one vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of passages each of the plurality of passages fanning outwardly from the passage and each of the plurality of passages being separated by one of a plurality of ribs located in the outer platform.

[0017] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum has a larger volume than each one of the plurality of passages.

[0018] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, cooling openings extending from the internal plenum to an external surface of the leading edge.

[0019] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, cooling openings extending from the internal plenum to any one of an external surface of the leading edge, matefaces of the outer platform, and a radially inwardly facing surface of the outer platform.

[0020] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum is located between the plurality of passages and an external surface of the outer platform.

[0021] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an area of each of the passages is equal.

[0022] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, each of the plurality of passages are connected to the internal plenum.

[0023] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the at least one vane is formed from a casting process.

[0024] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, upper surfaces and/or lower surfaces of the plurality of passages have trip strips extending therefrom.

[0025] Also disclosed is a core for forming a vane for a gas turbine engine, the vane including: an airfoil extending between an inner platform and an outer platform, the outer platform of the vane having a leading edge and a trailing edge; a passage located in the vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of passages each of the plurality of passages fanning outwardly from the passage and each of the plurality of passages being separated by one of a plurality of ribs located in the outer platform, the core including: a forward region that corresponds to the internal plenum; a plurality of fingers that correspond to the plurality of passages, the plurality of fingers extending from the forward region; and a plurality of openings located between the plurality of fingers, the plurality of openings corresponding to the plurality of ribs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0027] FIG. 1 is a schematic, partial cross-sectional view of a gas turbine engine in accordance with this disclosure;

[0028] FIG. 2 is a schematic view of a portion of a turbine section of the gas turbine engine;
[0029] FIG. 3 is a view of a turbine vane in accordance with the present disclosure;
[0030] FIG. 4 is a perspective view of an upper portion of the vane illustrated in FIG. 3 with a ceramic core;
[0031] FIG. 5 is a view along lines 5-5 in FIG. 3;
[0032] FIG. 6 is an opposite view of a portion of the vane illustrated in FIG. 5;
[0033] FIG. 7 is a perspective view of a core for forming a portion of the vane of the present disclosure;
[0034] FIG. 8 a perspective view of a portion of a vane of the present disclosure;
[0035] FIG. 9 is a view along lines 9-9 in FIG. 8;
[0036] FIG. 10 a perspective view of a portion of a vane of the present disclosure;
[0037] FIG. 11 is a view along lines 11-11 in FIG. 10; and
[0038] FIG. 12 is a view along lines 12-12 in FIG. 10.

DETAILED DESCRIPTION

[0039] A detailed description of one or more embodiments of the disclosed turbine vane and method of cooling are presented herein by way of exemplification and not limitation with reference to the FIGS.

[0040] FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 disclosed herein is provided as one non-limiting example of an engine the vane of the present disclosure may be used in. In other words, the turbine vane and method of cooling may be used in any suitable gas turbine engine and its use is not limited to the specific engine architectures illustrated in the attached FIGS. The gas turbine engine 20 illustrated in FIG. 1 may be referred to as a two-spool turbofan that generally incorporates a fan section 22, a compressor section 24, a combustor section 26 and a turbine section 28. Alternative engines might include other systems or features. The fan section 22 drives air along a bypass flow path B in a bypass duct, while the compressor section 24 drives air along a core flow path C for compression and communication into the combustor section 26 then expansion through the turbine section 28. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines including three-spool architectures.

[0041] The exemplary engine 20 generally includes a low speed spool 30 and a high speed spool 32 mounted for rotation about an engine central longitudinal axis A relative to an engine static structure 36 via several bearing systems 38. It should be understood that various bearing systems 38 at various locations may alternatively or additionally be provided, and the location of bearing systems 38 may be varied as appropriate to the application.

[0042] The low speed spool 30 generally includes an inner shaft 40 that interconnects a fan 42, a first or low pressure compressor 44 and a first or low pressure turbine 46. The inner shaft 40 is connected to the fan 42 through a speed change mechanism, which in exemplary gas turbine engine 20 is illustrated as a geared architecture 48 to drive the fan 42 at a lower speed than the low speed spool 30. It being understood that various embodiments of the present disclosure are applicable to engines that may or may not have the aforementioned geared architecture 48. The high speed spool 32 includes an outer shaft 50 that interconnects a second or high pressure compressor 52 and a second or high pressure turbine 54. A combustor 56 is arranged in exemplary gas turbine 20 between the high pressure compressor 52 and the high pressure turbine 54. A mid-turbine frame 57 of the engine static structure 36 is arranged generally between the high pressure turbine 54 and the low pressure turbine 46. The mid-turbine frame 57 further supports bearing systems 38 in the turbine section 28. The inner shaft 40 and the outer shaft 50 are concentric and rotate via bearing systems 38 about the engine central longitudinal axis A which is collinear with their longitudinal axes.

[0043] The core airflow is compressed by the low pressure compressor 44 then the high pressure

compressor **52**, mixed and burned with fuel in the combustor **56**, then expanded over the high pressure turbine **54** and low pressure turbine **46**. The mid-turbine frame **57** includes airfoils **59** which are in the core airflow path C. The turbines **46**, **54** rotationally drive the respective low speed spool **30** and high speed spool **32** in response to the expansion. It will be appreciated that each of the positions and configurations of the fan section **22**, compressor section **24**, combustor section **26**, turbine section **28**, and fan drive gear system **48** may be varied. For example, gear system **48** may be located aft of combustor section **26** or even aft of turbine section **28** or removed entirely, and fan section **22** may have different configurations and/or may be positioned forward or aft of the location of gear system **48**.

[0044] FIG. 2 illustrates a portion of the high pressure turbine (HPT) **54**. FIG. 2 also illustrates at least one high pressure turbine stage vane **70** that is located aft of a first one of a pair of turbine disks **72** and forward of the other one of the pair of turbine disks **72**. Each of the pair of turbine disks **72** having a plurality of turbine blades **74** secured thereto. The turbine blades **74** rotate proximate to blade outer air seals (BOAS) **76** which are located radially outward of tips of the turbine blades **74**.

[0045] The high pressure turbine stage vanes **70** are positioned circumferentially about the axis A of the engine. Hot gases from the combustor section **26** flow through the turbine in the direction of arrow **78**. Although a two-stage high pressure turbine is illustrated other high pressure turbines are considered to be within the scope of various embodiments of the present disclosure. Each of the vanes **70** have an airfoil **80** extending between an inner platform **82** and an outer platform **84**. The blade outer air seals (BOAS) **76** and the outer platform **84** are secured to a case **86** that circumferentially surrounds the high pressure turbine **54**.

[0046] The high pressure turbine (HPT) **54** is subjected to gas temperatures well above the yield capability of its material. In order to mitigate such high temperature detrimental effects, surface film-cooling is typically used to cool the blades and vanes of the high pressure turbine. Surface film-cooling is achieved by supplying cooling air from the cold backside through cooling holes drilled on the high pressure turbine components. Cooling holes are strategically designed and placed on the vane and turbine components in-order to maximize the cooling effectiveness and minimize the efficiency penalty.

[0047] In addition, internal cooling passageways and interconnecting cooling openings or crossovers are provided to allow for cooling air flow within the blades and vanes of the high pressure turbine. The outer platform **84** and the case **86** defines a cavity **88** which receives cooling air from a cooling air pipe **90** that is in fluid communication with the cavity **88**. Although only one cooling air pipe **90** is illustrated, a plurality of cooling air pipes may be positioned about a circumference of the case **86**.

[0048] Secured to the inner platform **82** is an inner air seal or full hoop inner seal **92**, which in conjunction with the inner platform **82** forms a vane inner diameter cavity **94**. The vane inner diameter cavity **94** is in fluid communication with a rotor cavity **96** via at least one opening **98** in the inner air seal or full hoop inner seal **92**.

[0049] In accordance with the present disclosure the outer platform **84** of the vane **70** is provided with a cooling scheme that provides uniform cooling to a surface of the outer platform **84**. In addition, the cooling design of the outer platform **84** employs a design that also improves the stiffness of a core used for forming the vane **70** of the present disclosure.

[0050] Referring now to at least FIGS. 1-12, a vane **70** in accordance with the present disclosure is illustrated. As mentioned above, each of the vanes **70** have an airfoil **80** extending between an inner platform **82** and an outer platform **84**. The airfoil **80** of the vane **70** has a leading edge **100** and a trailing edge **102**. The leading edge **100** encountering the hot gases from the combustor section before the trailing edge **102**. Similarly, the outer platform **84** of the vane **70** has a leading edge **104** and a trailing edge **106**. The leading edge **104** encountering the hot gases from the combustor section before the trailing edge **106**.

[0051] The inner platform **82** and the outer platform **84** are mounted into the engine structure at both radially inner and radially outer end of the airfoil **80**. As known, the airfoil **80** serves to redirect the products of combustion between turbine rotor stages.

[0052] As shown in FIG. 5, the airfoil **80** has cooling passages **108** extending therethrough. The cooling passages **108** are in fluid communication with a passage **110** in outer platform **84**. See at least FIG. 4. The passage **110** supplies cooling flow to passages **112** that provide cooling air forwardly toward the leading edge **104** of the outer platform **84**. As illustrated, multiple passages **112** are separated by fanned ribs **114** to improve passage flow fill and to maintain uniform flow per area. Flow exiting the cooling passages **112** is illustrated by arrows **116** in at least FIG. 4. In accordance with the present disclosure, all of the passages **112** connect into a manifold, internal plenum or bump-out region **118**. The manifold, internal plenum or bump-out region **118** has a larger volume than passages **112**. As illustrated in at least FIG. 4, the manifold, internal plenum or bump-out region **118** is located between the plurality of passages **112** and an external surface of the outer platform **84**.

[0053] In order to provide the cooling airflow in the direction of arrows **116**, cooling openings **120** are drilled into the vane **70** after it have been formed. The cooling openings **120** are drilled until they are in fluid communication with the manifold, internal plenum or bump-out region **118**.

[0054] Once the vane is fully formed the cooling openings **120** are formed or drilled into anyone of or all of an exterior surface of the leading edge **104**, matefaces **122** and a radially inwardly facing surface **124** of the outer platform **84**. In one embodiment, the manifold, internal plenum or bump-out region **118** may have a “U” shaped configuration such that it is located between the exterior surface of the leading edge **104** and the exterior surface of the matefaces **122** of the outer platform **84** and the passages **112**.

[0055] Referring now to FIG. 7, a perspective view of a core **126** for forming the forward section of outer platform **84**. As illustrated in at least FIG. 4 the core **126** is applied to a partially formed vane **70** so that the aforementioned passages **112**, ribs **114** and internal plenum or bump-out region **118** can be formed with the vane **70**. In other words, the vane **70** comprising at least the airfoil **80**, the inner platform **82** and at least a portion of the outer platform **84** is first formed. Thereafter and after these portions are formed the core **126** is applied to the partially formed vane **70**. The core **126** has a forward region **128** that corresponds to the manifold, internal plenum or bump-out region **118** and a plurality of fingers **130** that correspond to passages **112** and a plurality of openings **132** that correspond to the ribs **114**.

[0056] As is known in the related arts, the vane **70** is cast or is formed from a casting process, and typically utilizes a lost core molding technique. For example, a first portion of the vane without the passages **112**, ribs **114** and internal plenum or bump-out region **118** is formed by the lost core molding technique. Thereafter, the partially formed vane has the core **126** applied thereto and the passages **112**, ribs **114** and internal plenum or bump-out region **118** is formed by the lost core molding technique to provide a fully formed vane **70** in accordance with the present disclosure. As mentioned above, the core **126** will include spaces or openings **132** for the ribs **114**, and is solid at the location of the passages **112** and the manifold or bump out region **118** as well as passage **110**. After metal is cast around the core **126**, the core **126** is leached away, leaving the vane **70** as shown in the figures. Thus, the ribs **114** cast. The plurality of fingers **130** that correspond to passages **112** improve the core stiffness and provide a more producible design.

[0057] The openings **132** that correspond to the ribs **114** terminate short from the enlarged zone or manifold or bump out region **118** where cooling holes **120** are drilled to provide fluid flow for the cooling air.

[0058] The configuration of the fingers **130** on the platform core **126** allow the corresponding passages **112** to distribute cooling flow uniformly and towards the leading edge **104** of the outer platform **84**. The ribs **114** are placed to improve internal flow fill characteristics (i.e., mitigate regions of internal flow separation and recirculation). In an alternative configuration, the ribs **114**

could be positioned such that the flow over area is equality distributed along the passages **112**, which allows for more uniform metal temperature distribution on the platform **84**. In other words, the area of the passages **112** is equal. All the cavities **112** connect into the manifold, internal plenum or bump-out region **118**. The manifold, internal plenum or bump-out region **118** corresponds to the forward region **128** of the core **126** that is enlarged and improves the core **126** stiffness. This makes the core **126** less prone to breaking while keeping the design manufacturable through conventional casting process. Cooling holes **120** are also drilled into this enlarged plenum or bump out region **118** to get the flow out. The bump-out zone allows to holes **120** to be drilled from the surfaces of the outer platform **84** (e.g., leading edge **104**, radially inwardly facing surface **124**, and mateface sides **122**). As shown in FIG. 7, the openings **132** correspond to the ribs **114** that terminate short of the bump-out region or plenum **118** formed by area **128** of the core **126**.

Terminating of the ribs short of the bump-out region or plenum **118** facilitates a film hole **120** layout optimization regarding hole, row spacing and orientation. The cooling holes **120** could also be drilled into the individual passages to get cooling flow out where needed. As such, the cooling air flow is metered by drilled cooling holes **120** instead of a cast feature.

[0059] As mentioned above, turbine airfoils can be operating in a gas-path temperature far exceeding their melting point. To endure these temperatures, they must be cooled to an acceptable service temperature in order to maintain their integrity. In accordance with the present disclosure, the core **126** is designed to accommodate complex internal cooling passages to channel and direct the cooling air in a fan shaped configuration as illustrated in at least FIG. 4.

[0060] The fanned ribs **114** of the platform core **84** allow the passages **112** to distribute cooling flow uniformly and towards the leading edge **104**. The ribs **114** are placed to improve internal flow fill characteristics (i.e., mitigate regions of internal flow separation and recirculation). In another configuration, the ribs **114** could be positioned such that flow over area is equality distributed along the passages **112**, which allows for more uniform metal temperature distribution on the platform **84**. All cavities **112** connect into a manifold (bump-out region) **118** which also improves the core **126** stiffness. This makes the core **126** less prone to breaking while keeping the design manufacturable through conventional casting process.

[0061] FIGS. 5 and 6 are views illustrating ribs **114** and an upper surfaces **140** or lower surfaces **142** of passages **112**. As illustrated, the upper surfaces **140** and/or the lower surfaces **142** may be provided with trip strips **144** in order to enhance the convective cooling through passages **112**. As used herein, trip strips are protrusions that extend away from a surface such that the disrupt airflow passing past the trip strips in order to create eddies or swirling flow patterns of the cooling airflow in order to enhance the cooling provided by the cooling airflow. For example, the eddies or swirling flow patterns improves the cooling of portions of the vane **70** proximate to the eddies or swirling flow patterns of the cooling airflow.

[0062] While the vane **70** is shown as having a single airfoil **80** extending between the opposed platforms **82**, **84**, the present disclosure would also extend to vanes having a plurality of airfoils **80** connected to each platform **82**, **84**.

[0063] Although, FIG. 2 shows that the vane **70** may be used in the turbine section **28** in either or both of the high pressure and low pressure sections, the vane **70** may be used in other locations of the engine **20** in addition to or as an alternative to the turbine section **28**. Non limiting locations include, the compressor section (high and low) as well as another other section of the engine **20** that has a vane **70** or component needing a cooling air flow.

[0064] As used herein forward and aft are with reference to the core and bypass flow path from the fan section **22** (e.g., forward) towards the turbine section **28** (e.g., aft). In addition, the term radially is with reference to the engine central longitudinal axis A. For example, radially inward refers to a direction towards the engine central longitudinal axis A and radially outward refers to a direction away from the engine central longitudinal axis A. As such, radially inward platforms **82** are closer to the engine central longitudinal axis A than the radially outward platforms **84**, when the vane **70**

is installed in the gas turbine engine 20.

[0065] The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5%, or 2% of a given value.

[0066] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0067] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A vane for a gas turbine engine, comprising: an airfoil extending between an inner platform and an outer platform, the outer platform of the vane having a leading edge and a trailing edge; a passage located in the vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of internal passages each of the plurality of internal passages extending from the passage and fanning outwardly from the passage and each of the plurality of internal passages being separated by one of a plurality of ribs located in the outer platform.
2. The vane as in claim 1, wherein the internal plenum has a larger volume than each one of the plurality of passages.
3. The vane as in claim 1, further comprising: cooling openings extending from the internal plenum to an external surface of the leading edge.
4. The vane as in claim 1, further comprising: cooling openings extending from the internal plenum to any one of an external surface of the leading edge, matefaces of the outer platform, and a radially inwardly facing surface of the outer platform.
5. The vane as in claim 1, wherein the internal plenum is located between the plurality of passages and an external surface of the outer platform.
6. The vane as in claim 1, wherein an area of each of the plurality of internal passages is equal.
7. (canceled)
8. The vane as in claim 1, wherein the vane is a turbine vane.
9. The vane as in claim 1, wherein the vane is formed from a casting process.
10. The vane as in claim 1, wherein upper surfaces and/or lower surfaces of the plurality of internal passages have trip strips extending therefrom.
11. A gas turbine engine, comprising: a fan section; a compressor section; a combustor section; and a turbine section, the turbine section including at least one vane, the at least one vane comprising: an airfoil extending between an inner platform and an outer platform, the outer platform of the at least one vane having a leading edge and a trailing edge; a passage located in the at least one vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of

internal passages each of the plurality of internal passages extending from the passage and fanning outwardly from the passage and each of the plurality of internal passages being separated by one of a plurality of ribs located in the outer platform.

12. The gas turbine engine as in claim 11, wherein the internal plenum has a larger volume than each one of the plurality of internal passages.

13. The gas turbine engine as in claim 11, further comprising: cooling openings extending from the internal plenum to an external surface of the leading edge.

14. The gas turbine engine as in claim 11, further comprising: cooling openings extending from the internal plenum to any one of an external surface of the leading edge, matefaces of the outer platform, and a radially inwardly facing surface of the outer platform.

15. The gas turbine engine as in claim 11, wherein the internal plenum is located between the plurality of internal passages and an external surface of the outer platform.

16. The gas turbine engine as in claim 11, wherein an area of each of the plurality of internal passages is equal.

17. (canceled)

18. The gas turbine engine as in claim 11, wherein the at least one vane is formed from a casting process.

19. The gas turbine engine as in claim 11, wherein upper surfaces and/or lower surfaces of the plurality of internal passages have trip strips extending therefrom.

20. A core for forming a vane for a gas turbine engine, the vane including: an airfoil extending between an inner platform and an outer platform, the outer platform of the vane having a leading edge and a trailing edge; a passage located in the vane for providing cooling airflow to the leading edge; and an internal plenum located in the outer platform, the internal plenum being in fluid communication with the passage via a plurality of internal passages each of the plurality of internal passages extending from the passage and fanning outwardly from the passage and each of the plurality of internal passages being separated by one of a plurality of ribs located in the outer platform, the core comprising: a forward region that corresponds to the internal plenum; a plurality of fingers that correspond to the plurality of internal passages, the plurality of fingers extending from the forward region; and a plurality of openings located between the plurality of fingers, the plurality of openings corresponding to the plurality of ribs.
